IGCSE 0607 Practice Questions — E6.2

Area and perimeter

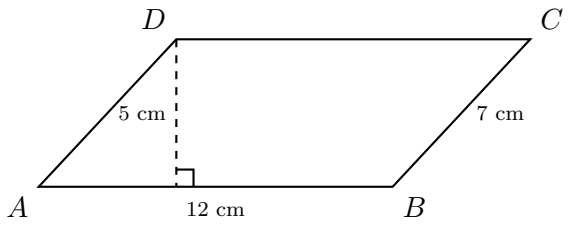
Total Questions: 18 — **Total Marks:** 126 Marks

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- **E6.2** Area and perimeter
  → **Paper 2 (Non-Calculator)** *(7 Qs)* · **Paper 4 (Calculator)** *(11 Qs)*

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1 The diagram shows a parallelogram $ABCD$ with $AB = 12$ cm and $BC = 7$ cm.
The perpendicular height from D to AB is 5 cm.



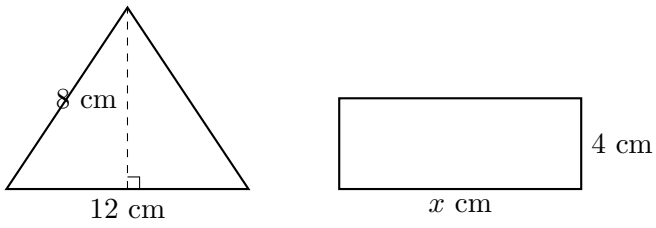
(a) Calculate the perimeter of parallelogram $ABCD$.

Perimeter = ... cm **[2]**

(b) Calculate the area of parallelogram $ABCD$.

Area = ... cm^2 **[2]**

2 An isosceles triangle and a rectangle are shown below.
The area of the triangle is equal to the area of the rectangle.



NOT TO SCALE

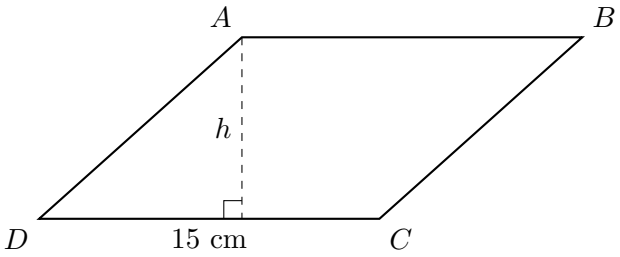
(a) Calculate the value of x.

$x =$ ..... **[2]**

(b) Calculate the perimeter of the isosceles triangle.

..... cm **[3]**

3 The diagram shows a parallelogram $ABCD$.
The area of the parallelogram is 120 cm^2 and its perimeter is 54 cm.



(a) Work out the value of h.

$h =$ ..... cm **[2]**

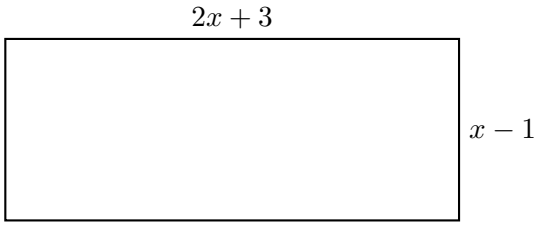
(b) Work out the length of AD.

$AD =$ ..... cm **[2]**

(c) Calculate the area of triangle ACD.

Area = ... cm^2 **[1]**

4 The diagram shows a rectangle.
All measurements are in centimetres.



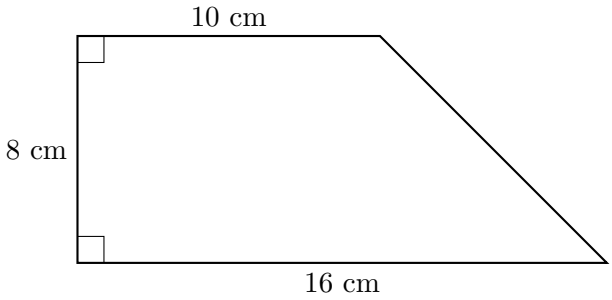
(a) The perimeter of the rectangle is 34 cm.
Find the value of x.

$x =$ ..... **[3]**

(b) Calculate the area of the rectangle.

..... cm^2 **[2]**

5 The diagram shows a right-angled trapezium.



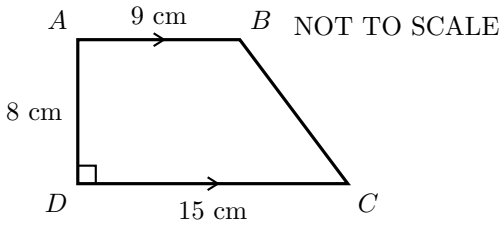
(a) Calculate the area of the trapezium.

..... cm^2 **[2]**

(b) Calculate the perimeter of the trapezium.

..... cm **[3]**

6 The diagram shows a trapezium $ABCD$ in which AB is parallel to DC and angle $ADC = 90°$.
$AB = 9$ cm, $DC = 15$ cm, and $AD = 8$ cm.



(a) Calculate the area of trapezium $ABCD$.

..... cm^2 **[2]**

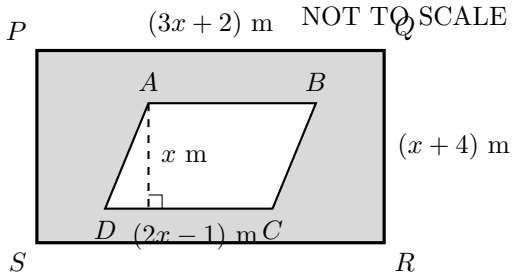
(b) Calculate the length of BC, and hence write down the perimeter of trapezium $ABCD$.

$BC =$ ..... , Perimeter = ... **[3]**

(c) Point E lies on DC such that $ABCE$ is a parallelogram.
Calculate the area of triangle ADE.

..... cm^2 **[2]**

7 The diagram shows a rectangular garden $PQRS$ of length $(3x + 2)$ m and width $(x + 4)$ m. Inside the garden, a flowerbed in the shape of a parallelogram $ABCD$ has base $(2x - 1)$ m and perpendicular height x m. The shaded region surrounding the flowerbed is covered with grass.



(a) Write down and simplify an expression, in terms of x, for the perimeter of the rectangular garden $PQRS$.

..... m **[2]**

(b) Show that the area of the grass, A m^2, is given by $A = x^2 + 15x + 8$.

[3]

(c) The area of the grass is 62 m^2.

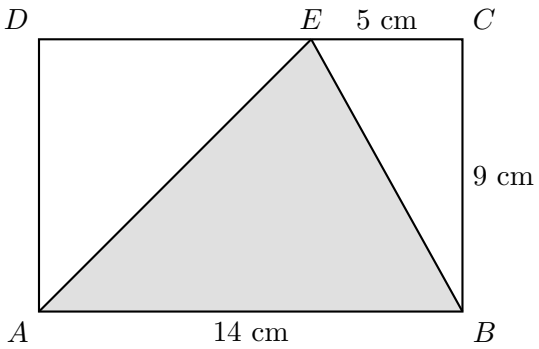
(i) Write down a quadratic equation in terms of x and solve it to calculate the value of x.

$x =$ ..... **[3]**

(ii) Calculate the area of the flowerbed $ABCD$.

..... m^2 **[1]**

8 $ABCD$ is a rectangle with length 14 cm and width 9 cm.
The point E lies on the edge CD such that $CE = 5$ cm.



NOT TO SCALE

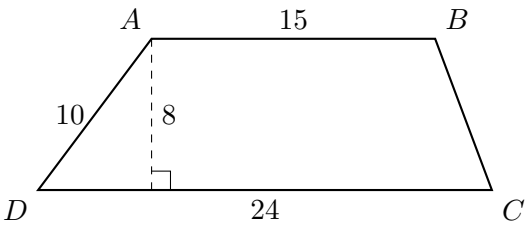
(a) Calculate the area of the shaded triangle ABE.

..... cm^2 **[2]**

(b) Calculate the perimeter of the shaded triangle ABE.

..... cm **[4]**

9 The diagram shows a trapezium $ABCD$ with AB parallel to DC.
The perpendicular height of the trapezium is 8 cm.
All measurements are in centimetres.



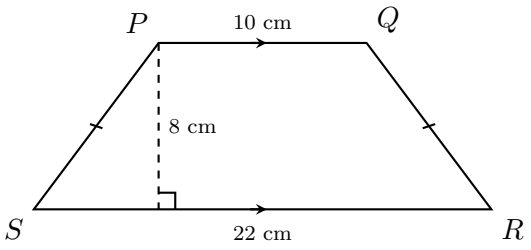
(a) Calculate the area of the trapezium.

..... cm^2 **[2]**

(b) Calculate the perimeter of the trapezium.

..... cm **[4]**

10 The diagram shows an isosceles trapezium $PQRS$ with PQ parallel to SR.
$PQ = 10$ cm, $SR = 22$ cm, and the perpendicular height of the trapezium is 8 cm.



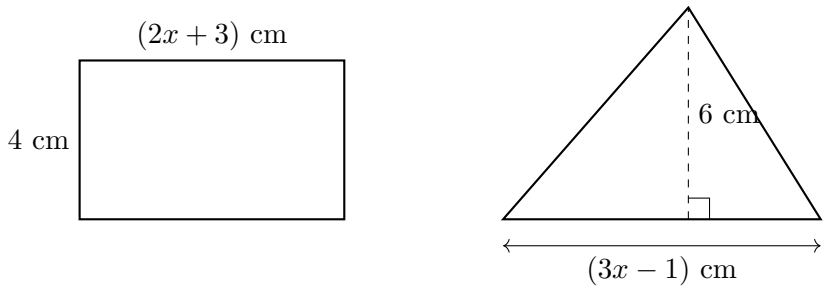
(a) Calculate the area of the trapezium $PQRS$.

Area = ... cm^2 **[2]**

(b) Calculate the perimeter of the trapezium $PQRS$.

Perimeter = ... cm **[4]**

11 The diagram shows a rectangle and a triangle.
The area of the rectangle is equal to the area of the triangle.



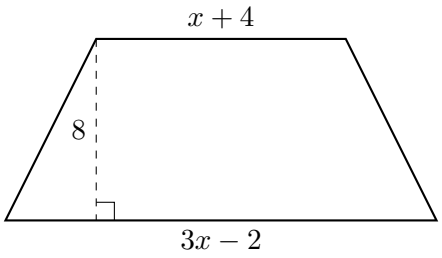
(a) Form an equation in x and solve it to find the value of x.

$x =$ ..... **[4]**

(b) Work out the perimeter of the rectangle.

..... cm **[2]**

12 The diagram shows a trapezium.
All measurements are in centimetres.



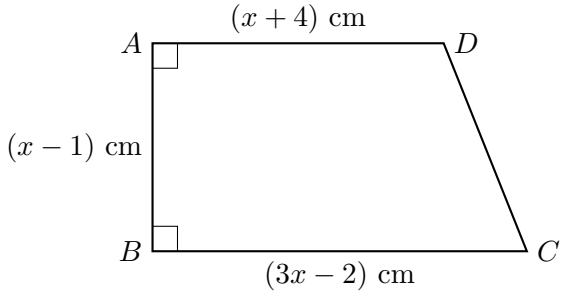
(a) The area of the trapezium is 120 cm^2.
Work out the value of x.

$x =$ ..... **[3]**

(b) The trapezium is an isosceles trapezium, with the two slanted sides being equal in length.
Calculate the perimeter of the trapezium.

Perimeter = ... cm **[4]**

13 A trapezium $ABCD$ has right angles at A and B.
The side AD is parallel to BC.
$AD = (x + 4)$ cm, $BC = (3x - 2)$ cm and $AB = (x - 1)$ cm.



NOT TO SCALE

(a) The area of the trapezium is 20 cm^2.

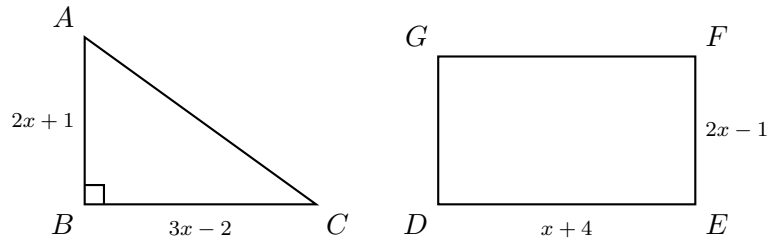
(i) Show that $2x^2 - x - 21 = 0$.

[3]

(ii) Calculate the perimeter of the trapezium.

..... cm **[5]**

14 The diagram shows a right-angled triangle ABC and a rectangle $DEFG$.
All lengths are in centimetres.
In triangle ABC, $AB = (2x + 1)$ cm and $BC = (3x - 2)$ cm.
In rectangle $DEFG$, $DE = (x + 4)$ cm and $EF = (2x - 1)$ cm.



(a) The area of triangle ABC is equal to the area of rectangle $DEFG$.
Show that $2x^2 - 15x + 6 = 0$.

[3]

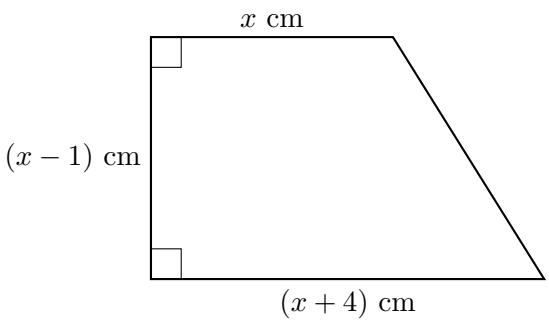
(b) Solve the equation $2x^2 - 15x + 6 = 0$.
Give your answers correct to 2 decimal places.

$x =$ ..... , $x =$ ..... **[3]**

(c) Calculate the perimeter of rectangle $DEFG$.

Perimeter = ... cm **[2]**

15 The diagram shows a right-angled trapezium with parallel sides of length x cm and $(x + 4)$ cm.
The height of the trapezium is $(x - 1)$ cm.
The area of the trapezium is 40 cm^2.



(a) Show that $x^2 + x - 42 = 0$.

[3]

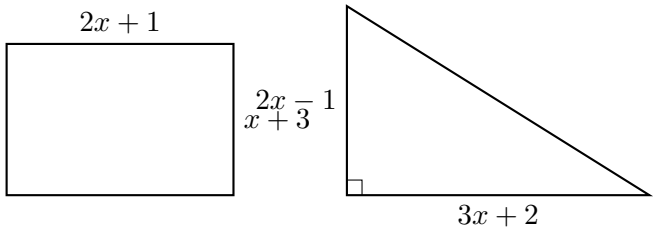
(b) Solve the equation $x^2 + x - 42 = 0$ to find the value of x.

$x =$ ..... **[2]**

(c) Calculate the perimeter of the trapezium.

..... cm **[4]**

16 A rectangle and a right-angled triangle are shown.
All measurements are in centimetres.



(a) The area of the rectangle is equal to the area of the right-angled triangle.
Show that $2x^2 - 13x - 8 = 0$.

[4]

(b) Solve the equation $2x^2 - 13x - 8 = 0$.
Show all your working and give your answers correct to 2 decimal places.

$x =$ ..... , $x =$ ..... **[4]**

(c) Calculate the perimeter of the rectangle.

Perimeter = ... cm **[2]**

17 A piece of wire of length 60 cm is cut into two pieces.
One piece is bent to form a square.
The other piece is bent to form a rectangle where the length is exactly twice the width.
Let the width of the rectangle be y cm.

(a) Write down an expression, in terms of y, for the side length of the square.

..... cm **[2]**

(b) The total area enclosed by the square and the rectangle is 113 cm^2.
Show that $17y^2 - 180y + 448 = 0$.

[3]

(c) Solve the equation.

$$17y^2 - 180y + 448 = 0$$

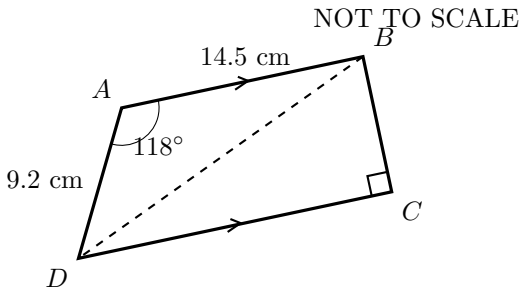
Show all your working.

$y =$ ..... , $y =$ ..... **[3]**

(d) Given that the side length of the square must be greater than the width of the rectangle, calculate the area of the square.

..... cm^2 **[2]**

18 The diagram shows a trapezium $ABCD$ in which AB is parallel to DC and angle $BCD = 90°$. $AB = 14.5$ cm, $AD = 9.2$ cm, and angle $BAD = 118°$.



(a) Calculate the length of diagonal BD, giving your answer correct to 3 significant figures.

$BD =$ ..... cm **[3]**

(b) Show that angle $BDC = 23.3°$, correct to 1 decimal place.

[2]

(c) Calculate the perimeter of trapezium $ABCD$, giving your answer correct to 3 significant figures.

Perimeter = ... cm **[3]**

(d) Calculate the area of trapezium $ABCD$, giving your answer correct to 3 significant figures.

Area = ... cm^2 **[2]**